

README – Desert Camp Security System

Project Description

This project implements a security system designed to protect a desert camp during night hours. The system detects intruders approaching the protected area and activates visual and audible alarms. An authorized guard can reset the alarm using an IR remote control.

Hardware Components

- Arduino Uno
- Ultrasonic Distance Sensor
- Push Button
- LED
- Buzzer
- IR Receiver
- IR Remote Control
- Servo Motor
- LCD 16x2 Display
- Potentiometer

Main Features

- System arming and disarming using a push button.
- Intruder detection using an ultrasonic sensor.
- Visual alarm using an LED.
- Audible alarm using a buzzer.
- Warning flag activation using a servo motor.
- Alarm reset using an IR remote control.
- Real-time system status display on an LCD screen.

Pin Configuration

Component	Arduino Pin	Component	Arduino Pin
Button	D2	Button	D2
LCD RS	D3	LCD RS	D3
Ultrasonic TRIG	D4	Ultrasonic TRIG	D4
Ultrasonic ECHO	D5	Ultrasonic ECHO	D5

IR Receiver D6	IR Receiver D6
Buzzer D7	Buzzer D7
LED D8	LED D8
Servo Motor D9	Servo Motor D9
LCD E A0	LCD E A0
LCD D4 A1	LCD D4 A1
LCD D5 A2	LCD D5 A2
LCD D6 A3	LCD D6 A3
LCD D7 A4	LCD D7 A4

System Workflow

1. The user arms the system using the push button.
2. The ultrasonic sensor continuously monitors the protected area.
3. When an object is detected closer than 30 cm, the alarm is activated.
4. The LED, buzzer, servo motor, and LCD warning message are activated.
5. The authorized guard resets the alarm using the IR remote control.
6. The system returns to monitoring mode.

Files Included

- Arduino source code
- Project report
- README file